

Hybrid OCR System for Handwritten Text Recognition using Tesseract and TrOCR

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MOHANA PRIYA R - (230701191)

PRATHEBA D - (230701239)

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Project titled “**Hybrid Handwritten Text Recognition System using OCR and Transformer-Based Deep Learning**” is the bonafide work of “**MOHANA PRIYA R (230701191), PRATHEBA (230701239)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. M.E.Malathy Ph.D.,

HEAD OF THE DEPARTMENT

Professor

Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai - 602 105.

SIGNATURE

Dr. M. Rakesh Kumar., M.E., Ph.D.,

SUPERVISOR

Assistant Professor

Department of Computer Science
and Engineering,
Rajalakshmi Engineering
College, Chennai-602 105.

Submitted to Mini Project Viva-Voce Examination held on _____

Internal Examiner

External Examiner

ABSTRACT

Hybrid Handwritten Text Recognition System using OCR and

Transformer-Based Deep Learning" is an advanced solution designed to convert handwritten text into accurate and editable digital content. The system addresses the challenges associated with digitizing handwritten documents, such as variability in writing styles, noise in images, and low-quality inputs. By integrating traditional Optical Character Recognition (OCR) using Tesseract and deep learning techniques through transformer-based TrOCR models, the system ensures both speed and high accuracy in text extraction.

The proposed system incorporates image preprocessing techniques, including grayscale conversion, noise removal, and adaptive thresholding, to enhance input quality and improve recognition performance. It analyzes handwritten inputs and intelligently selects between OCR methods to deliver optimal results for both neat and complex handwriting styles. The application is developed using a user-friendly Gradio-based web interface, allowing users to upload handwritten images and instantly receive digital text output.

The performance of the system is evaluated based on accuracy, processing speed, and robustness across different handwriting samples. The hybrid approach enhances reliability by combining the strengths of rule-based and deep learning models. By providing a scalable, efficient, and accessible solution, this system aims to simplify document digitization, reduce manual effort, and improve productivity. This innovative framework can be widely applied in educational, administrative, and professional domains, enabling seamless conversion of handwritten content into digital form while ensuring accuracy and usability.

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MOHANA PRIYA R 2116230701191

PRATHEBA D 2116230701239

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LIST OF ABBREVIATIONS

S. No	ABBR	Expansion
1	AI	Artificial Intelligence
2`	API	Application Programming Interface
3	AJAX	Asynchronous JavaScript and XML
4	ASGI	Asynchronous Server Gateway Interface
5	AWT	Abstract Window Toolkit
6	BC	Block Chain
7	CSS	Cascading Style Sheet
8	DFD	Data Flow Diagram
9	DSS	Digital Signature Scheme
10	GB	Gradient Boosting
11	JSON	JavaScript Object Notation
12	ML	Machine Learning
13	RF	Random Forest
14	SQL	Structure Query Language
15	SVM	Support Vector Machine

INTRODUCTION

1.1 GENERAL

In the modern digital era, the need to convert handwritten content into digital format has become increasingly important across various domains such as education, business, healthcare, and administration. Despite the rapid advancement of digital technologies, handwritten documents are still widely used for note-taking, record-keeping, and communication. However, managing and processing handwritten data manually is time-consuming, error-prone, and inefficient. This creates a strong demand for intelligent systems that can accurately recognize and convert handwritten text into machine-readable format.

Handwritten Text Recognition (HTR) is a challenging task in the field of computer vision and pattern recognition due to the variability in individual writing styles, differences in character shapes, and issues such as noise, lighting variations, and image quality. Traditional Optical Character Recognition (OCR) systems, such as Tesseract, perform well on printed text and structured handwriting but often struggle with complex or irregular handwritten inputs. To overcome these limitations, recent advancements in deep learning, particularly transformer-based models, have significantly improved the accuracy and robustness of text recognition systems.

This project presents a **Hybrid Handwritten Text Recognition System** that combines the strengths of traditional OCR techniques and modern deep learning approaches. The system integrates Tesseract OCR for fast and efficient recognition of neat handwriting, along with a transformer-based TrOCR model for handling more complex and unstructured handwritten text. Additionally, image preprocessing

techniques such as grayscale conversion, noise removal, and adaptive thresholding are applied to enhance image quality and improve recognition performance.

The proposed system is implemented using Python and provides a user-friendly interface through a Gradio-based web application. Users can upload handwritten images and obtain digital text output in real time. This approach not only improves accuracy but also ensures flexibility by allowing users to choose between different recognition methods based on their requirements.

Overall, this project aims to develop an efficient, accurate, and scalable solution for handwritten text recognition. It reduces manual effort, enhances productivity, and enables easy storage, editing, and retrieval of handwritten information in digital form, making it highly beneficial for real-world applications.

1.2 OBJECTIVE

The main objective of this project is to design and develop an efficient Handwritten Text Recognition (HTR) system that converts handwritten text from images into accurate and editable digital text. The system aims to improve recognition accuracy across different handwriting styles by integrating both traditional Optical Character Recognition (OCR) and advanced deep learning techniques. It utilizes Tesseract OCR for fast and reliable extraction of text from neat handwriting, while transformer-based TrOCR models are used to effectively handle complex and unstructured handwriting. Another key objective is to enhance the quality of input images through preprocessing techniques such as grayscale conversion, noise removal, and adaptive thresholding to improve overall recognition performance. The project also focuses on reducing the manual effort required for typing handwritten content and providing a simple, user-friendly interface for uploading and processing images. Additionally, it ensures quick and real-time text extraction, making the system practical and efficient for users. The solution is

designed to be adaptable for various real-world applications such as education, document management, and administrative tasks. It aims to maintain a balance between speed and accuracy while offering scalability for future enhancements. Overall, the objective is to develop a reliable, intelligent, and efficient system for handwritten text digitization.

1.2 EXISTING SYSTEM

The existing systems for text recognition are primarily based on traditional Optical Character Recognition (OCR) techniques, such as Tesseract, which are mainly designed for printed text and structured documents. These systems perform well when the input text is clear, neatly written, and properly aligned, but they face significant challenges when dealing with handwritten content. Variations in handwriting styles, differences in character shapes, and irregular spacing often lead to inaccurate results. Additionally, existing OCR systems are highly sensitive to image quality issues such as noise, poor lighting, and low resolution, which further reduces their efficiency.

Most traditional systems lack advanced preprocessing techniques, making it difficult to handle complex or distorted inputs. Furthermore, they do not incorporate modern deep learning approaches that can adapt and learn from diverse handwriting patterns. As a result, their performance is limited when applied to real-world handwritten documents. Many existing solutions also lack user-friendly interfaces and real-time processing capabilities, making them less accessible for general users. Therefore, there is a need for a more advanced and reliable system that can overcome these limitations and provide accurate handwritten text recognition.

CHAPTER 2

LITERATURE SURVEY

"Handwritten Text Recognition using Deep Learning Techniques" [1] (2023) by Sharma R., et al. explores the application of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) for recognizing handwritten text from images. The study focuses on automatic feature extraction and sequence modeling to improve recognition accuracy. By training on large datasets, the model demonstrates significant improvements over traditional OCR systems in handling complex handwriting styles. However, the approach requires high computational resources and large amounts of labeled data, which may limit its practical implementation in resource-constrained environments.

"Optical Character Recognition System using Tesseract OCR Engine" [2] (2022) by Patel S., et al. presents a system based on the Tesseract OCR engine for extracting text from images. The research highlights the efficiency and speed of Tesseract in recognizing printed and neatly written text. It also discusses preprocessing techniques such as binarization and noise removal to improve performance. While the system performs well for structured inputs, its major limitation is poor accuracy when dealing with irregular or cursive handwriting, making it less suitable for real-world handwritten documents.

"Transformer-Based Optical Character Recognition with TrOCR" [3] (2021) by Li M., et al. introduces a transformer-based model that combines vision and language understanding for end-to-end text recognition. The model uses self-attention mechanisms to capture contextual relationships within text, leading to improved accuracy in recognizing both printed and handwritten content. This approach reduces the need for complex pipelines and achieves state-of-the-art results. However, the model requires significant computational power and initial setup time, which can be a

challenge for real-time or low-resource applications.

"Image Preprocessing Techniques for Improved OCR Accuracy" [4] (2023) by Singh A., et al. emphasizes the importance of preprocessing steps such as grayscale conversion, noise reduction, and adaptive thresholding in enhancing OCR performance. The study demonstrates that proper preprocessing significantly improves text recognition accuracy by enhancing image clarity and reducing distortions. Despite its effectiveness, preprocessing alone cannot fully overcome challenges posed by highly complex or distorted handwriting styles.

"Hybrid Approach for Handwritten Text Recognition" [5] (2024) by Verma K., et al. proposes a hybrid system that combines traditional OCR methods with deep learning models to achieve better accuracy and efficiency. The study highlights that integrating multiple techniques helps balance speed and performance, making the system more adaptable to various handwriting styles. However, the complexity of integrating multiple models and maintaining system efficiency remains a key challenge.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

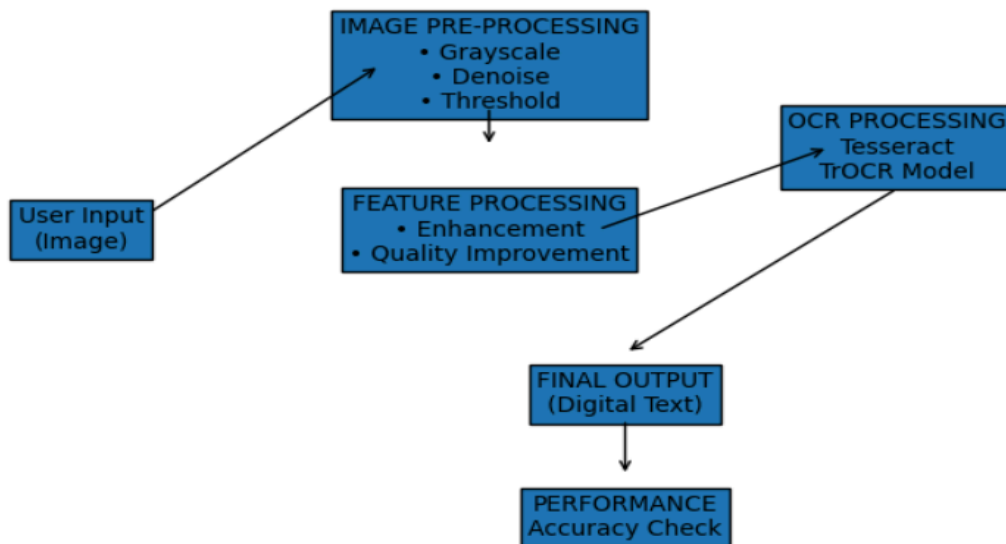
The proposed system is a **Hybrid Handwritten Text Recognition System** designed to efficiently convert handwritten text from images into accurate digital text using a combination of traditional OCR and advanced deep learning techniques. This system aims to overcome the limitations of existing OCR methods by integrating Tesseract OCR with a transformer-based TrOCR model, thereby improving both speed and accuracy in text recognition.

The system accepts handwritten images as input through a user-friendly interface and processes them using a series of image preprocessing techniques such as grayscale conversion, noise removal, and adaptive thresholding. These preprocessing steps enhance the quality of the input image and improve the overall recognition performance. Based on the selected method, the system then applies either Tesseract OCR for faster processing of neat handwriting or the TrOCR deep learning model for more complex and unstructured handwriting.

The hybrid approach ensures flexibility by allowing the system to adapt to different handwriting styles and image conditions. The integration of deep learning enables the system to learn and recognize patterns more effectively, while the OCR component ensures quick and efficient text extraction. The output is displayed in a digital text format, which can be easily copied, edited, or stored for future use.

The entire system is implemented using Python and deployed through a Gradio-based web application, making it accessible and easy to use without requiring advanced technical knowledge. Overall, the proposed system provides a reliable, scalable, and efficient solution for handwritten text recognition, suitable for real-world applications in education, administration, and document digitization.

Fig 3.1: System Architecture



3.1 DEVELOPMENTAL ENVIRONMENT

3.1.1 HARDWARE REQUIREMENTS

The system requires a standard computer or laptop with a minimum of 4 GB RAM to ensure smooth execution of the application. A processor such as Intel i3 or higher is recommended for efficient performance. Adequate storage space is needed to install required software and libraries. A basic input device like a keyboard and mouse is necessary for user interaction. Additionally, a stable internet connection is required for downloading models and dependencies during setup.

3.1.2 SOFTWARE REQUIREMENTS

The system requires Python as the primary programming language for implementation and execution of the application. Libraries such as OpenCV, Pillow, and NumPy are used for image preprocessing and manipulation. Pytesseract is used for traditional OCR, while the Transformers library with PyTorch is used to implement the TrOCR deep learning model. Gradio is used to develop a simple and interactive user interface for the application. Additionally, a

code editor such as Visual Studio Code or PyCharm is required for development and testing purposes..

3.2 DESIGN OF THE ENTIRE SYSTEM

3.2.1 ACTIVITY DIAGRAM

The activity diagram represents the workflow of the proposed Handwritten Text Recognition system. The process begins with the user uploading a handwritten image through the interface. The system then performs image preprocessing steps such as grayscale conversion, noise removal, and thresholding to enhance image quality. After preprocessing, the user selects the OCR method based on their requirement.

If the user selects Tesseract OCR, the system processes the image using a traditional OCR approach suitable for neat handwriting. If the user selects the TrOCR model, the system applies a deep learning-based approach for better accuracy in complex handwriting. The outputs from both methods are then converted into digital text format. Finally, the extracted text is displayed to the user through the interface, and the process ends.

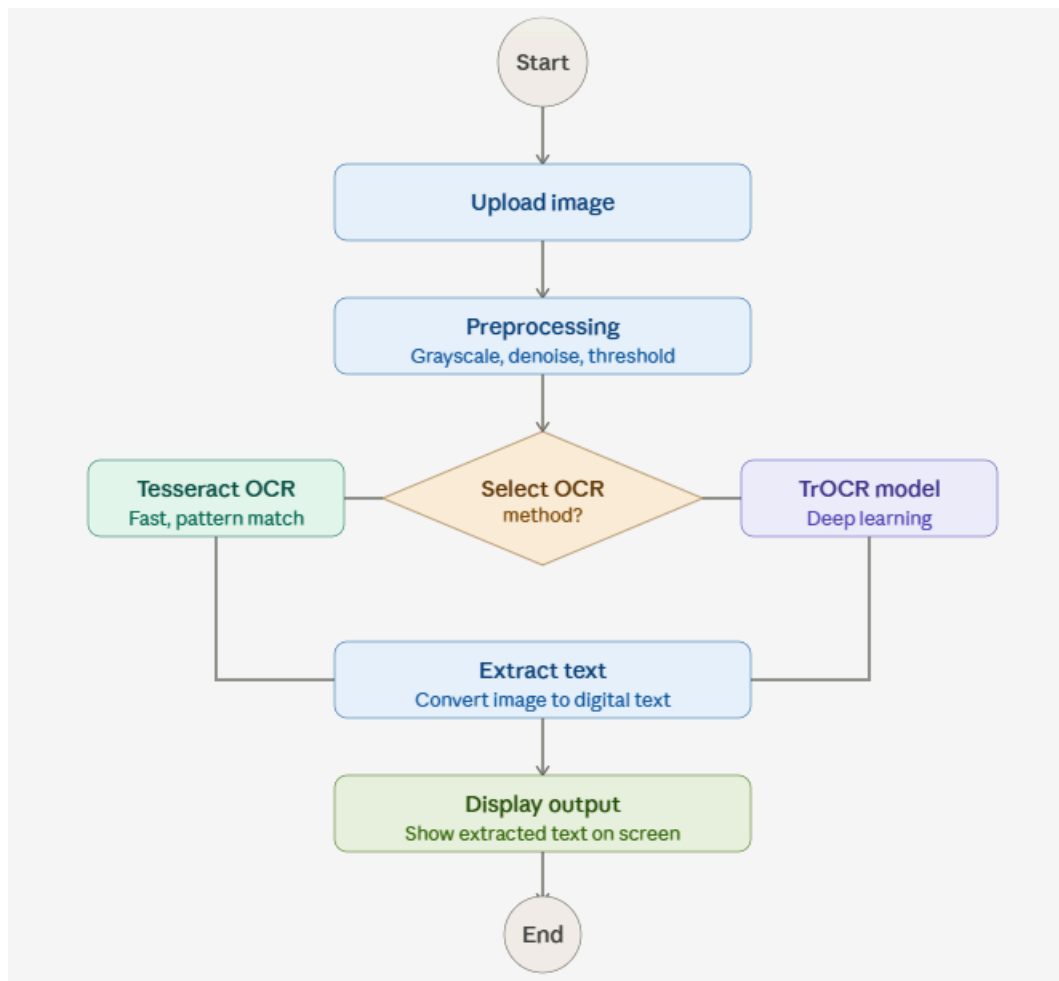


Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates how data moves through the Handwritten Text Recognition system and how it is processed at each stage. The process begins with the user providing a handwritten image as input through the interface. This input data is then passed to the preprocessing module, where image enhancement techniques such as grayscale conversion, noise removal, and thresholding are applied to improve image quality. The processed image is then forwarded to the OCR processing unit, where the user-selected method—either Tesseract OCR or the TrOCR deep learning model—is applied to extract text from the image.

During this stage, the system analyzes the image and converts the visual handwritten content into machine-readable text. The extracted text is then passed to the output module, where it is formatted and displayed to the user through the interface. The system ensures smooth data flow between each module, maintaining efficiency and accuracy throughout the process. Additionally, the system may store temporary data during processing to improve performance. Overall, the Data Flow Diagram represents a structured flow of input, processing, and output, ensuring that handwritten data is effectively transformed into digital text.

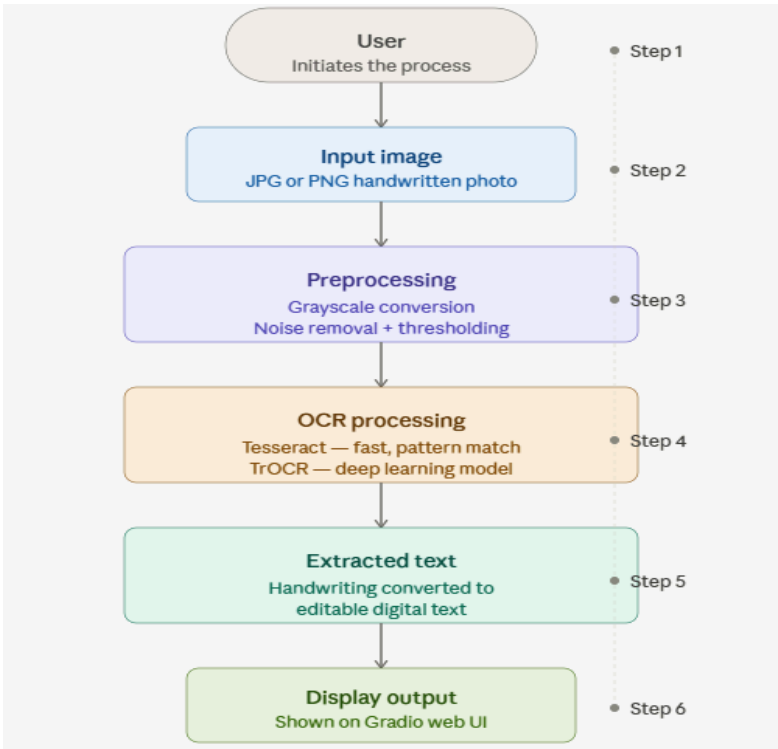


Fig 3.3:Data Flow Diagram

3.3 STATISTICAL ANALYSIS

Statistical analysis plays an important role in evaluating the performance and efficiency of the Handwritten Text Recognition system. In this project, the system’s performance is analyzed using key metrics such as accuracy, precision, recall, and processing time. Accuracy measures the correctness of the extracted

text compared to the actual handwritten content, while precision and recall help in evaluating how effectively the system identifies and recognizes characters without errors.

The system is tested using different handwritten samples with varying writing styles, sizes, and image qualities to ensure robustness. The results obtained from both Tesseract OCR and TrOCR models are compared to understand their performance differences. Tesseract generally provides faster results with moderate accuracy for neat handwriting, whereas the TrOCR model achieves higher accuracy for complex and unstructured handwriting but requires more processing time.

Additionally, statistical observations are made to analyze the impact of preprocessing techniques such as noise removal and thresholding on recognition accuracy. The results indicate that proper preprocessing significantly improves the overall performance of the system. The analysis also helps in identifying limitations and areas for improvement, ensuring that the system can be further enhanced for better accuracy and efficiency. Overall, statistical analysis provides a clear understanding of the system's effectiveness and supports the validation of the proposed approach.

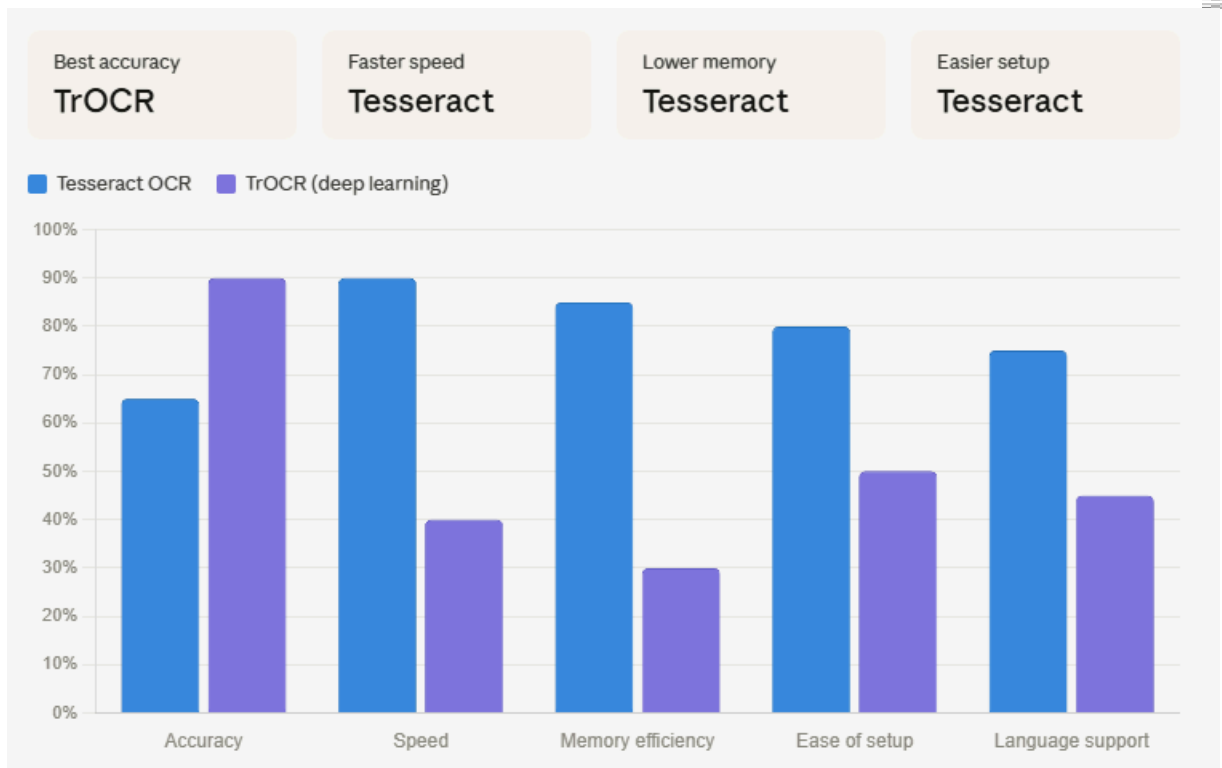


Fig 3.4 : Comparison Graph

The comparison graph evaluates two OCR methods used in this project — Tesseract OCR and TrOCR — across five key parameters: accuracy, speed, memory efficiency, ease of setup, and language support. In terms of accuracy, TrOCR scores significantly higher at 90 out of 100 compared to Tesseract's 65, because TrOCR is a deep learning model trained by Microsoft on millions of handwriting samples, making it far better at understanding messy or cursive handwriting.

However, when it comes to speed, Tesseract dominates with a score of 90, while TrOCR scores only 40, since TrOCR must load a large 1.3 GB neural network model which takes considerable time. In memory efficiency, Tesseract again leads with 85 against TrOCR's 30, as Tesseract is a lightweight traditional OCR engine that does not require heavy computing resources. Regarding ease of setup, Tesseract scores 80 compared to TrOCR's 50, because Tesseract only requires a simple system installation while TrOCR requires downloading large model files and deep learning libraries.

Finally, in language support, Tesseract scores 75 against TrOCR's 45, since Tesseract supports over 100 languages through language packs while TrOCR is primarily trained on English text. Overall, the graph clearly shows that TrOCR is the better choice when accuracy is the priority, while Tesseract is the preferred option when speed, memory, and simplicity are more important — which is why this project provides both methods as options to the user.

CHAPTER 4

MODULE DESCRIPTION

The workflow for the proposed system is designed to ensure a structured and efficient process for recognizing Handwritten text Image .It consists of the following sequential steps:

4.1 SYSTEM ARCHITECTURE

4.1.1 USER INTERFACE DESIGN

The user interface of the Handwritten Text Recognition system was developed to provide a simple, interactive, and user-friendly experience for converting handwritten text into digital text. The frontend is built using the Gradio framework, which allows easy integration between machine learning models and web applications.

The interface consists of an image upload section where users can either drag and drop a handwritten image or manually upload it from the local system. Once the image is uploaded, the preview of the handwritten image is displayed on the left side of the interface. On the right side, an output text box is provided to display the extracted digital text generated by the OCR model.

4.1.2 BACK END INFRASTRUCTURE

The backend of the project is responsible for performing image processing, model execution, OCR prediction, and returning the extracted text to the user interface. It is implemented using Python and integrates multiple libraries such as:

- OpenCV for image preprocessing
- TensorFlow and Keras for deep learning model execution

- Tesseract OCR for traditional OCR
- TrOCR for transformer-based handwritten text recognition

The backend workflow begins when the user uploads an image. The image is first preprocessed using grayscale conversion, Gaussian blur, thresholding, and normalization techniques to improve recognition quality. After preprocessing, the selected OCR model is executed.

4.2 DATA COLLECTION AND PREPROCESSING

4.2.1 Dataset and Data Labelling

This project employs the IAM Handwriting Dataset, a well-known benchmark dataset for handwritten text recognition tasks.

It includes:

- Words, lines and paragraphs written by hand
- generalization -> multiple writers
- For this project:
- Initially: word-level dataset (IAM Words)
- Final approach: line-level data set (IAM Lines)

4.2.2. Data Preprocessing

Preprocessing is critical to make the handwritten text readable by the model.

Steps Applied

1. **Grayscale Conversion**

- Converts RGB image → single channel

2. **Binarization (Otsu Thresholding)**

- Separates text from background

3. **Noise Handling & Augmentation**

- Rotation ($\pm 5^\circ$)
- Gaussian blur
- Random noise

4. **Aspect Ratio Preservation**

- Maintains natural handwriting proportions

5. **Resizing**

- Final size: **32 × 512**

6. **Padding**

- Center padding applied for uniform input

7. **Normalization**

- Pixel values scaled to **[0,1]**

4.2.3 Feature Selection

The system employs automatic feature extraction with deep learning instead of manual feature engineering.

Features Extracted:

1. Strokes pattern.
2. Character Forms
3. Flow of writing
4. Spacing between characters

Technique:

1. CNN (Convolutional Neural Networks)
2. Spatial features extraction
3. No need for hand-crafted features

4.2.4 Classification and Model Selection

The architecture used is a CRNN (Convolutional Recurrent Neural Network).

Components:

1. Convolutional Neural Network Layer

Extract visual features from images

2. RNN Layer (BiLSTM)

Captures sequential dependencies in text

3. CTC Loss (Connectionist Temporal Classification)

allows recognition without character alignment

Why this model?

- Supports variable length texts
- Suitable for hand writing recognition
- Removes segmentation requirement

4.2.5 Performance Evaluation and Optimization

To evaluate the model, the following metrics are used:

1. CER (Character Error Rate)

- Measures character-level mistakes

$$\text{CER} = (\text{Edit Distance}) / (\text{Total Character})$$

$$\text{CER} = \frac{\text{Total Characters Edit Distance}}{\text{Total Characters}}$$

2. WER (Word Error Rate)

- Measures word-level errors

$$\text{WER} = (\text{Word Errors}) / (\text{Total Words})$$

3. Observations:

- Lower CER → better character recognition
- Lower WER → better sentence understanding

4.2.6 Model Deployment

Deployment Strategy:

- Stored locally (for now)
- Can be:
 - Shared across systems
 - Loaded in backend (Flask/FastAPI)
 - Converted to mobile format later

Future Options:

- Convert to:
 - TensorFlow Lite (for mobile)
 - REST API service

4.2.7 Centralized Server and Database

Current Setup:

- **Local system**

Future Architecture:

- **Backend server (Flask / FastAPI)**
- **Database:**
 - **Stores:**
 - **Uploaded images**
 - **Recognized text**
 - **User history**

Benefits:

- **Centralized processing**
- **Scalable architecture**
- **Multi-user support**

4.3 SYSTEM WORK FLOW

Step-by-Step Workflow

1. **User uploads image (paragraph)**

2. Image is sent to backend
3. **Line segmentation** is applied
4. Each line is:
 - Preprocessed
 - Passed into CRNN model
5. Model predicts text using **CTC decoding**
6. Output is:
 - Cleaned
 - Post-processed
7. Final text is displayed to user

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The Handwritten Text Recognition (HTR) system was implemented using deep learning, image processing, and computer vision techniques to recognize handwritten text from document images. The project was developed using Python with TensorFlow, Keras, OpenCV, and NumPy libraries.

Initially, the IAM Words dataset was used, but later the IAM Lines dataset was adopted to support recognition of complete handwritten lines instead of isolated words. Image preprocessing techniques such as grayscale conversion, thresholding, resizing, normalization, and padding were applied to improve image quality and maintain uniform input dimensions.

The recognition model was built using a Convolutional Recurrent Neural Network (CRNN) architecture. CNN layers were used to extract handwriting features, while Bidirectional LSTM layers learned sequential text patterns. A Connectionist Temporal Classification (CTC) layer was integrated to enable sequence prediction without manual character segmentation.

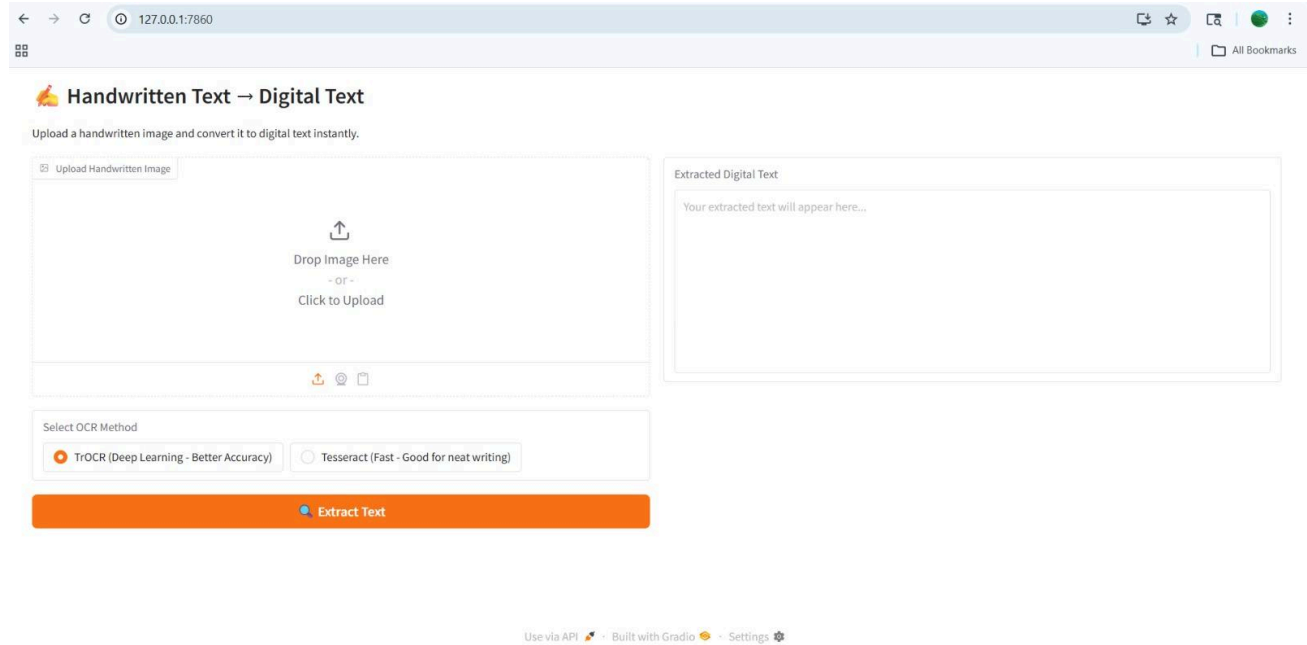
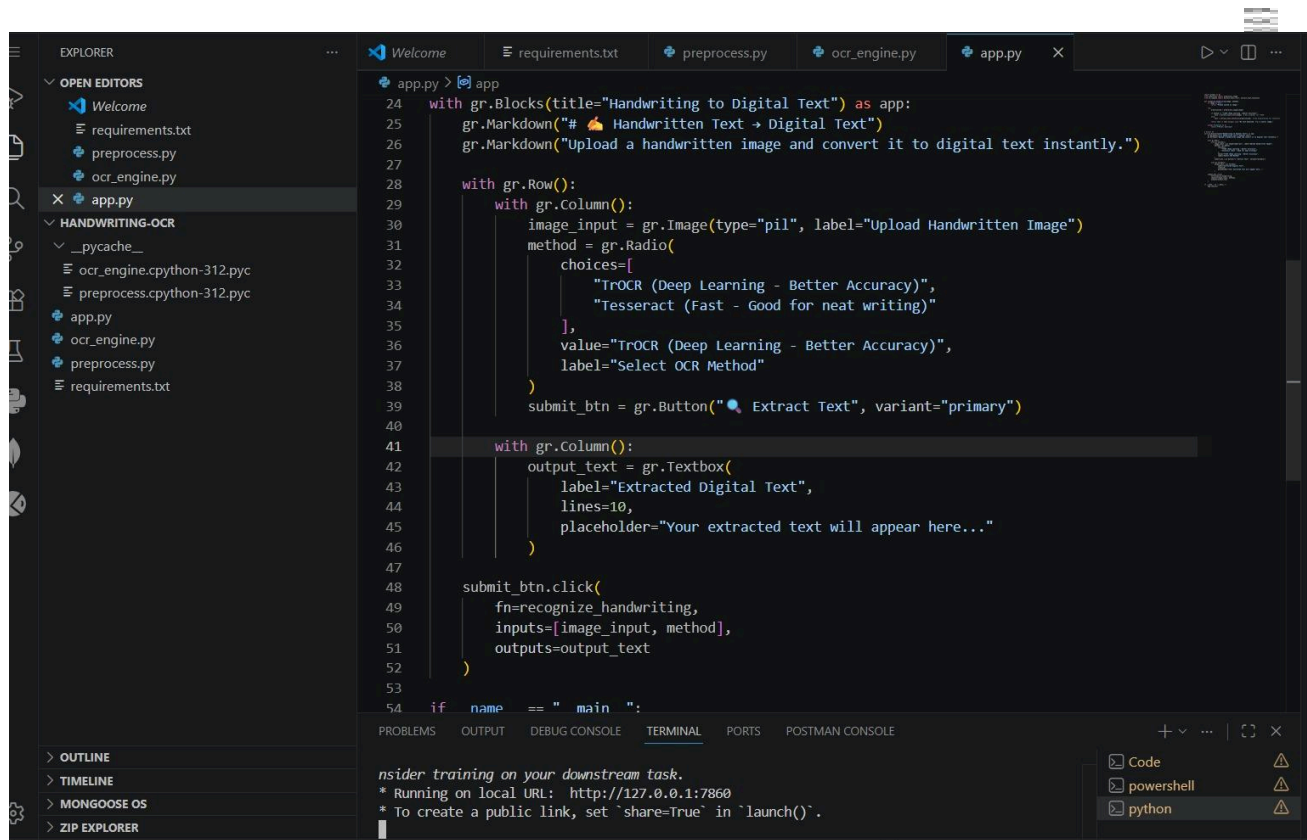
Data augmentation techniques including rotation, noise addition, and blurring were used during training to improve model robustness. Beam Search decoding was implemented during prediction for better accuracy.

5.2 OUTPUT SCREENSHOTS

The output of this Handwritten Text Recognition (HTR) project is the successful conversion of handwritten text images into editable and machine-readable digital text. The system accepts handwritten input images through image upload or camera capture, preprocesses the image using techniques such as grayscale conversion, Gaussian blur, thresholding, and segmentation, and then passes the processed image to the CRNN deep learning model for recognition.

The final output generated by the model is displayed as recognized text corresponding to the handwritten content present in the input image. The project is capable of identifying handwritten characters, words, and sentences by extracting spatial and sequential features using CNN and LSTM layers. The recognized text can then be stored, edited, searched, or used for further digital processing.

In addition to text prediction, the system also demonstrates intermediate outputs during preprocessing stages, such as grayscale images, binary images, and segmented text regions, which help in improving recognition accuracy and understanding the internal workflow of the model. The project therefore provides both functional handwritten text conversion and a complete demonstration of image processing and deep learning techniques used in modern OCR systems.



🔥 Handwritten Text → Digital Text

Upload a handwritten image and convert it to digital text instantly.

Upload Handwritten Image

MY PRACTICAL BOOKHAND

Based on a seventh century AD English half-uncial writing hand that features round letterforms coupled via extended lower serifs. The writing is characterized by great elegance and evenness. I find it very easy to render in a relaxed and easy manner. This exemplar was rendered using a Manuscript brand calligraphy fountain pen with an extra fine edged nib and Waterman bottled brown ink on Clairefontaine paper. It was freely written at everyday writing speed.



Select OCR Method

☒ TrOCR (Deep Learning - Better Accuracy)

☐ Tesseract (Fast - Good for neat writing)

Extract Text

Extracted Digital Text

Your extracted text will appear here...

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The Handwritten Text Recognition (HTR) project was successfully developed to convert handwritten text images into machine-readable digital text using image processing and deep learning techniques. The system combines preprocessing methods such as grayscale conversion, Gaussian blur, thresholding, and segmentation with a CRNN (Convolutional Recurrent Neural Network) architecture consisting of CNN and LSTM layers. CNN was used for extracting spatial features from handwritten images, while LSTM handled sequential character prediction. CTC loss was applied to recognize variable-length text without requiring exact character alignment.

The project demonstrated how computer vision and machine learning can be integrated to solve real-world problems involving handwritten data digitization. The implementation using Keras and TensorFlow simplified model development, training, and evaluation while providing flexibility for future improvements. The preprocessing pipeline improved image clarity and reduced noise, which helped enhance recognition performance.

One of the major achievements of this project was understanding the complete workflow of an HTR system, including image acquisition, preprocessing, feature extraction, sequence modeling, and prediction decoding. The project also provided practical exposure to neural network architectures, dataset handling, model training, and performance optimization techniques.

Although the model performs effectively for handwritten text recognition, there is still scope for enhancement in terms of accuracy and robustness for complex handwriting

styles. Future improvements can include larger datasets, transformer-based architectures such as TrOCR, attention mechanisms, advanced augmentation techniques, and cloud deployment for real-time applications.

Overall, this project successfully demonstrates the practical application of image processing, computer vision, deep learning, and software engineering concepts in building an intelligent handwritten text recognition system.

6.2 FUTURE ENHANCEMENT

Future enhancements for this project include improving recognition accuracy using larger and more diverse handwritten datasets, implementing advanced preprocessing techniques, and integrating transformer-based models such as TrOCR for better sequence understanding. The system can also be enhanced with attention mechanisms, real-time cloud deployment, multilingual handwriting support, mobile application integration, and improved decoding methods such as beam search to make the model faster, more scalable, and more accurate for real-world applications.

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